Solutions to Axler's *Linear Algebra Done Right*, 4th Edition

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8B Generalized Eigenspace Decomposition

Exercise 8B1

Define $T \in \mathcal{L}(\mathbb{C}^2)$ by $T(w,z) = (-z,w)$. Find the generalized eigenspaces corresponding to the distinct eigenvalues of T.

Solution. If $T(w,z) = \lambda(w,z)$ for $\lambda \in \mathbb{C}$ and nonzero $(w,z) \in \mathbb{C}^2$, then $-z = \lambda w$ and $w = \lambda z$. Together, these equations imply $(\lambda^2 + 1)z = 0$. Since $z \neq 0$, for otherwise $w = 0$, we must have $\lambda^2 + 1 = 0$, whence $\lambda = \pm i$. If $\lambda = i$, then $-z = iw$, so $E(i,T) = \operatorname{span}((i,1))$. A similar argument shows that $E(-i,T) = \operatorname{span}((-i,1))$. Since $E(\pm i,T) \subseteq G(\pm i,T)$, each generalized eigenspace must have dimension at least 1. Moreover, the multiplicities must sum to 2 by 8.25, so each generalized eigenspace has dimension exactly 1. By 2.39,

$$G(i,T) = \operatorname{span}((i,1)), \quad G(-i,T) = \operatorname{span}((-i,1)).$$

■

Exercise 8B2

Suppose $T \in \mathcal{L}(V)$ is invertible. Prove that $G(\lambda,T) = G\left(\frac{1}{\lambda}, T^{-1}\right)$ for every $\lambda \in \mathbb{F}$ with $\lambda \neq 0$.

Solution. Observe that $T^{-1} - \frac{1}{\lambda}I = -\frac{1}{\lambda}T^{-1}(T - \lambda I)$. Moreover, $T^{-1}(T - \lambda I) = (T - \lambda I)T^{-1} = I - \lambda T^{-1}$. Then for all positive integers k,

$$\left(T^{-1} - \frac{1}{\lambda}I\right)^k = \left(-\frac{1}{\lambda}T^{-1}(T - \lambda I)\right)^k = \left(-\frac{1}{\lambda}\right)^k T^{-k}(T - \lambda I)^k.$$

Since $\lambda \neq 0$, the operator $(-\frac{1}{\lambda})^k T^{-k}$ is injective, and therefore $\operatorname{null}(T^{-1} - \frac{1}{\lambda}I)^k = \operatorname{null}(T - \lambda I)^k$. By 8.19, the result follows. ■

Exercise 8B3

Suppose $T \in \mathcal{L}(V)$. Suppose $S \in \mathcal{L}(V)$ is invertible. Prove that T and $S^{-1}TS$ have the same eigenvalues with the same multiplicities.

Solution. Note that

$$(S^{-1}TS - \lambda I)^k = (S^{-1}(T - \lambda I)S)^k = S^{-1}(T - \lambda I)^k S.$$

Because S^{-1} is injective, $S^{-1}(T-\lambda I)^k Sv = 0$ if and only if $(T-\lambda I)^k Sv = 0$, and therefore $\operatorname{null}(S^{-1}TS-\lambda I)^k = S^{-1}(\operatorname{null}(T - \lambda I)^k)$ for all positive integers k. For $k = 1$, we have $E(\lambda, S^{-1}TS) = S^{-1}(E(\lambda,T))$. Since S^{-1} is injective, $E(\lambda, S^{-1}TS) \neq \{0\}$ if and only if $E(\lambda,T) \neq \{0\}$, so T and $S^{-1}TS$ have the same eigenvalues. Taking $k = \dim V$ and using 8.20 yields $G(\lambda, S^{-1}TS) = S^{-1}(G(\lambda,T))$. Note that $S^{-1}|_{G(\lambda,T)} \in \mathcal{L}(G(\lambda,T), V)$ is an injective linear map whose range is $G(\lambda, S^{-1}TS)$. By the fundamental theorem of linear maps,

$$\dim G(\lambda,T) = \dim \operatorname{null} S^{-1}|_{G(\lambda,T)} + \dim \operatorname{range} S^{-1}|_{G(\lambda,T)} = \dim G(\lambda, S^{-1}TS),$$

so each eigenvalue has the same multiplicity for T as for $S^{-1}TS$. ■

Exercise 8B4

Suppose $\dim V \geq 2$ and $T \in \mathcal{L}(V)$ is such that $\operatorname{null} T^{\dim V-2} \neq \operatorname{null} T^{\dim V-1}$. Prove that T has at most two distinct eigenvalues.

Solution. Let $n = \dim V$. Because $\operatorname{null} T^{n-2} \neq \operatorname{null} T^{n-1}$, the contrapositive of 8.2 gives $\operatorname{null} T^k \neq \operatorname{null} T^{k+1}$ for all $k \leq n-2$. Then all the inclusions in the chain up to $\operatorname{null} T^{n-1}$ in 8.1 must be strict:

$$\{0\} = \operatorname{null} T^0 \subsetneq \operatorname{null} T^1 \subsetneq \cdots \subsetneq \operatorname{null} T^{n-1}.$$

Consequently, the dimension of each successive null space increases by at least one, so $\dim \operatorname{null} T^{n-1} \geq n-1$. Because $\operatorname{null} T \neq \{0\}$, 5.7 gives that 0 is an eigenvalue of T. By 8.1 and 8.20, $\operatorname{null} T^{n-1} \subseteq \operatorname{null} T^n = G(0,T)$, so $\dim G(0,T) \geq \dim \operatorname{null} T^{n-1} \geq n-1$. Let $0, \lambda_2, \ldots, \lambda_m$ be the distinct eigenvalues of T. By 8.12 and 1.45, $G(0,T) + G(\lambda_2,T) + \cdots + G(\lambda_m,T)$ is a direct sum, so by 3.94 and 2.37,

$$\dim G(0,T) + \dim G(\lambda_2,T) + \cdots + \dim G(\lambda_m,T) \leq \dim V = n.$$

Since each dimension must be at least 1, $(n-1) + (m-1) \leq n$, so $m \leq 2$. ∎

Exercise 8B5

Suppose $T \in \mathcal{L}(V)$ and 3 and 8 are eigenvalues of T. Let $n = \dim V$. Prove that $V = (\operatorname{null} T^{n-2}) \oplus (\operatorname{range} T^{n-2})$.

Manual Remark. *In fact, if T has m distinct nonzero eigenvalues, then $V = \operatorname{null} T^{n-m} \oplus \operatorname{range} T^{n-m}$.*

Solution. By 5.12, $n \geq 2$ since T has at least two distinct eigenvalues, 3 and 8. We claim that $\operatorname{null} T^{n-2} = \operatorname{null} T^{n-1}$. If 0 is not an eigenvalue of T, then 5.7 yields $\operatorname{null} T^0 = \operatorname{null} T^1 = \{0\}$, so 8.2 gives $\operatorname{null} T^{n-2} = \operatorname{null} T^{n-1}$. If 0 is an eigenvalue of T, then $0, 3, 8$ are three distinct eigenvalues, and the contrapositive of Exercise 8B4 gives $\operatorname{null} T^{n-2} = \operatorname{null} T^{n-1}$. In either case, we have $\operatorname{null} T^{n-2} = \operatorname{null} T^n$ by 8.2. Applying Exercise 8A9 with $m = n-2$ yields $\operatorname{range} T^{n-2} = \operatorname{range} T^{n-1}$, so $\operatorname{range} T^{n-2} = \operatorname{range} T^n$ by Exercise 8A7. Therefore, 8.4 gives

$$V = \operatorname{null} T^n \oplus \operatorname{range} T^n = \operatorname{null} T^{n-2} \oplus \operatorname{range} T^{n-2}.$$ ∎

Exercise 8B6

Suppose $T \in \mathcal{L}(V)$ and λ is an eigenvalue of T. Explain why the exponent of $z-\lambda$ in the factorization of the minimal polynomial of T is the smallest positive integer m such that $(T-\lambda I)^m|_{G(\lambda,T)} = 0$.

Solution. Let $n = \dim V$. Let p be the minimal polynomial of T. By 5.27(a), λ is a zero of p, so applying 4.6 repeatedly gives $p(z) = (z-\lambda)^k q(z)$ for some positive integer k and some $q \in \mathcal{P}(\mathbb{F})$ with $q(\lambda) \neq 0$. Let $G = G(\lambda,T)$ and $U = \operatorname{range}(T-\lambda I)^n$. By 8.4, $V = G \oplus U$, and both G and U are invariant under T by 5.18. Because $(T-\lambda I)^n|_G = 0$ by 8.20, there exists a smallest positive integer m such that $(T-\lambda I)^m|_G = 0$. We claim that $m = k$.

$\boxed{\leq}$ By 5.29, the minimal polynomial r of $T|_G$ divides $(z-\lambda)^m$, so by 4.13 if $\mathbb{F} = \mathbb{C}$ or 4.16 if $\mathbb{F} = \mathbb{R}$, we have $r(z) = (z-\lambda)^{m'}$ for some $m' \leq m$. Because λ is an eigenvalue of T, we have $\{0\} \neq E(\lambda,T) \subseteq G$. It follows that $r \neq 1$, so $m' \geq 1$. The minimality of m gives $m' = m$, so the minimal polynomial of $T|_G$

is $(z-\lambda)^m$. By 5.31, r also divides p. If $m > k$, then $(z-\lambda)^{m-k}$ would divide q, contradicting $q(\lambda) \neq 0$. Therefore, $m \leq k$.
$\boxed{\geq}$ Note that $\operatorname{null}(T-\lambda I) \subseteq G$ and $G \cap U = \{0\}$, so $(T-\lambda I)|_U$ is injective. For any $u \in U$, we have $(T-\lambda I)^k q(T)u = 0$, where $q(T)u \in U$. By injectivity, this implies $q(T)u = 0$. Let $s(z) = (z-\lambda)^m q(z)$. Then $s(T)u = (T-\lambda I)^m q(T)u = (T-\lambda I)^m 0 = 0$. On G, we have $(T-\lambda I)^m|_G = 0$, so $s(T)|_G = q(T)(T-\lambda I)^m|_G = 0$ as well. Therefore, $s(T) = 0$ on all of V. By 5.29, p divides s, so $m \geq k$, whence $m = k$. ■

Exercise 8B7

Suppose $T \in \mathcal{L}(V)$ and λ is an eigenvalue of T with multiplicity d. Prove that $G(\lambda, T) = \operatorname{null}(T-\lambda I)^d$.
Axler Note. *If $d < \dim V$, then this exercise improves 8.20.*

Solution. Let $n = \dim V$ and let $G = G(\lambda, T)$. Define $N = (T-\lambda I)|_G$. By 8.23, $\dim G = d$.
$\boxed{\subseteq}$ By 8.20, $G = \operatorname{null}(T-\lambda I)^n$, which is invariant under T by 5.18 applied to $(z-\lambda)^n$, and $N^n = 0$. Thus, $N \in \mathcal{L}(G)$, so $N^d = 0$ by 8.16. Hence, $G \subseteq \operatorname{null}(T-\lambda I)^d$.
$\boxed{\supseteq}$ Since λ is an eigenvalue of T, we have $\{0\} \neq E(\lambda, T) \subseteq G$, so $d \geq 1$. Hence, $\operatorname{null}(T-\lambda I)^d \subseteq G$ by 8.19. Therefore, $G = \operatorname{null}(T-\lambda I)^d$. ■

Exercise 8B8

Suppose $T \in \mathcal{L}(V)$ and $\lambda_1, \ldots, \lambda_m$ are the distinct eigenvalues of T. Prove that

$$V = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_m, T)$$

if and only if the minimal polynomial of T equals $(z-\lambda_1)^{k_1} \cdots (z-\lambda_m)^{k_m}$ for some positive integers $k_1, \ldots, k_m$.
Axler Note. *The case $\mathbb{F} = \mathbb{C}$ follows immediately from 5.27(b) and the generalized eigenspace decomposition (8.22); thus this exercise is interesting only when $\mathbb{F} = \mathbb{R}$.*

Solution.
$\boxed{\Rightarrow}$ By 3.94 and 2.42, taking bases of all the $G(\lambda_j, T)$ gives a basis of V consisting of generalized eigenvectors of T. By Exercise 8A11, the minimal polynomial p of T is $p(z) = (z-\mu_1)\cdots(z-\mu_M)$ for $\mu_1, \ldots, \mu_M \in \mathbb{F}$. By 5.27(a), $\{\mu_1, \ldots, \mu_M\} = \{\lambda_1, \ldots, \lambda_m\}$, so each λ_j appears with some positive exponent k_j. Grouping together the factors corresponding to the same λ_j gives the desired form $(z-\lambda_1)^{k_1} \cdots (z-\lambda_m)^{k_m}$.
$\boxed{\Leftarrow}$ By Exercise 8A11, V has a basis of generalized eigenvectors of T. Since each generalized eigenvector belongs to some $G(\lambda_j, T)$, the span of the union of the $G(\lambda_j, T)$ equals V, which gives

$$V = \sum_{j=1}^{m} G(\lambda_j, T).$$

By 8.12 and 1.45, this sum is direct. ■

Exercise 8B9

Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Prove that there exist $D, N \in \mathcal{L}(V)$ such that $T = D + N$, the operator D is diagonalizable, N is nilpotent, and $DN = ND$.

Solution. Let $n = \dim V$ and let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues of T. By the generalized eigenspace decomposition, $V = G_1 \oplus \cdots \oplus G_m$, where $G_j = G(\lambda_j, T)$. Define D by $D(v_1 + \cdots + v_m) = \lambda_1 v_1 + \cdots + \lambda_m v_m$ for $v_k \in G_k$. Since the sum is direct, D is well-defined and linear. Define $N = T - D$. Since concatenating bases of all the G_j gives a basis of eigenvectors of D by 3.94 and 2.42, it follows that D is diagonalizable by 5.55.

Suppose $u \in G_k$ for some k. By 8.22(a), G_k is invariant under T. Then $Nu = (T-D)u = Tu - \lambda_k u \in G_k$. It follows that $N^n u = (T - \lambda_k I)^n u = 0$ by 8.20, so $N^n = 0$ on each G_k. Moreover, $Tu \in G_k$, so $DTu = \lambda_k Tu = T(\lambda_k u) = TDu$, which gives $DT = TD$ on each G_k. This gives $DN = D(T - D) = DT - D^2 = TD - D^2 = (T - D)D = ND$ on each G_k. Since V is the direct sum of the G_k, it follows that N is nilpotent and $DN = ND$ on V. ■

Exercise 8B10

Suppose V is a complex inner product space, $e_1, \dots, e_n$ is an orthonormal basis of V, and $T \in \mathcal{L}(V)$. Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of T, each included as many times as its multiplicity. Prove that

$$|\lambda_1|^2 + \cdots + |\lambda_n|^2 \le \|Te_1\|^2 + \cdots + \|Te_n\|^2.$$

Axler Note. *See the comment after* Exercise 7A5.

Solution. By Schur's theorem, there exists an orthonormal basis $f_1, \dots, f_n$ of V such that the matrix A of T with respect to this basis is upper triangular. By 5.41, 8.31, and 8.25, the diagonal entries $A_{1,1}, \dots, A_{n,n}$ are the eigenvalues $\lambda_1, \dots, \lambda_n$ of T rearranged in some order. Observe that $Tf_k = \sum_{j \le k} A_{j,k} f_j$ for each k, so

$$\|Tf_k\|^2 = \sum_{j \le k} |A_{j,k}|^2 \ge |A_{k,k}|^2$$

by 6.24. By Exercise 7A5 applied to both bases $e_1, \dots, e_n$ and $f_1, \dots, f_n$, we have

$$\sum_{m=1}^{n} \|Te_m\|^2 = \sum_{m=1}^{n} \|Tf_m\|^2 \ge \sum_{m=1}^{n} |A_{m,m}|^2 = \sum_{m=1}^{n} |\lambda_m|^2.$$ ■

Exercise 8B11

Give an example of an operator on $\mathbb{C}^4$ whose characteristic polynomial equals $(z-7)^2(z-8)^2$.

Solution. Define $T \in \mathcal{L}(\mathbb{C}^4)$ by $T(z_1, z_2, z_3, z_4) = (7z_1, 7z_2, 8z_3, 8z_4)$. Then $\mathcal{M}(T)$ with respect to the standard basis is diagonal with diagonal entries $7, 7, 8, 8$. By 5.41, the eigenvalues of T are $7, 8$, each with multiplicity 2 by 8.31. Hence, the characteristic polynomial of T is $(z-7)^2(z-8)^2$. ■

Exercise 8B12

Give an example of an operator on $\mathbb{C}^4$ whose characteristic polynomial equals $(z-1)(z-5)^3$ and whose minimal polynomial equals $(z-1)(z-5)^2$.

Solution. Let e_1, e_2, e_3, e_4 be the standard basis of $\mathbb{C}^4$. Consider the operator $T \in \mathcal{L}(\mathbb{C}^4)$ defined by $T(z_1, z_2, z_3, z_4) = (z_1, 5z_2 + z_3, 5z_3, 5z_4)$. Then the matrix of T with respect to the standard basis is

$$\mathcal{M}(T) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 1 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 5 \end{pmatrix}.$$

By 5.41, the eigenvalues of T are $1, 5$, with 1 having multiplicity 1 and 5 having multiplicity 3 by 8.31. Therefore, the characteristic polynomial of T is $(z-1)(z-5)^3$. Let p be the minimal polynomial of T. By 8.30, 5.27(a), and uniqueness of the factorization of the minimal polynomial, we have $p(z) = (z-1)(z-5)^b$ for some $b \in \{1, 2, 3\}$. Observe that $(T - I)(T - 5I)e_3 = 4e_2 \neq 0$. However, $(T - 5I)^2 e_1 = 16e_1$ and $(T - 5I)^2 e_k = 0$ for $k = 2, 3, 4$. Moreover, $(T - I)e_1 = 0$, so $(T - I)(T - 5I)^2 = 0$. Hence, $b = 2$ and the minimal polynomial of T is $p(z) = (z-1)(z-5)^2$. ■

Exercise 8B13

Give an example of an operator on $\mathbb{C}^4$ whose characteristic and minimal polynomials both equal $z(z-1)^2(z-3)$.

Solution. Let e_1, e_2, e_3, e_4 be the standard basis of $\mathbb{C}^4$. Consider the operator $T \in \mathcal{L}(\mathbb{C}^4)$ defined by $T(z_1, z_2, z_3, z_4) = (0, z_2 + z_3, z_3, 3z_4)$. Then the matrix of T with respect to the standard basis is

$$\mathcal{M}(T) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

By 5.41, the eigenvalues of T are $0, 1, 3$, with 0 and 3 having multiplicity 1 and 1 having multiplicity 2 by 8.31. Therefore, the characteristic polynomial of T is $z(z-1)^2(z-3)$. Let p be the minimal polynomial of T. By 8.30, 5.27(a), and uniqueness of the factorization of the minimal polynomial, we have $p(z) = z(z-1)^b(z-3)$ for some $b \in \{1, 2\}$. Since $T(T - I)(T - 3I)e_3 = -2e_2 \neq 0$, we have $b = 2$, so $p(z) = z(z-1)^2(z-3)$. ■

Exercise 8B14

Give an example of an operator on $\mathbb{C}^4$ whose characteristic polynomial equals $z(z-1)^2(z-3)$ and whose minimal polynomial equals $z(z-1)(z-3)$.

Solution. Let $T \in \mathcal{L}(\mathbb{C}^4)$ be defined by $T(z_1, z_2, z_3, z_4) = (0, z_2, z_3, 3z_4)$. Then $\mathcal{M}(T)$ with respect to the standard basis is diagonal with diagonal entries $0, 1, 1, 3$. By 5.41, the eigenvalues of T are $0, 1, 3$, with 0

and 3 having multiplicity 1 and 1 having multiplicity 2 by 8.31. Therefore, the characteristic polynomial of T is $z(z-1)^2(z-3)$. Since $\mathcal{M}(T)$ is diagonal, T is diagonalizable, so the minimal polynomial of T has distinct linear factors by 5.62, and hence the minimal polynomial of T is $z(z-1)(z-3)$ by 5.27(a). ■

Exercise 8B15

Let T be the operator on $\mathbb{C}^4$ defined by $T(z_1, z_2, z_3, z_4) = (0, z_1, z_2, z_3)$. Find the characteristic polynomial and the minimal polynomial of T.

Solution. Let e_1, e_2, e_3, e_4 be the standard basis of $\mathbb{C}^4$. Observe that $Te_k = e_{k+1}$ for $k = 1, 2, 3$ and $Te_4 = 0$, so $T^4 = 0$, and hence T is nilpotent. By 8.17(a), 0 is the only eigenvalue of T, and $\dim \operatorname{null} T^4 = \dim \mathbb{C}^4 = 4$, so the multiplicity of 0 is 4 by 8.23, and hence the characteristic polynomial of T is z^4. Moreover, 8.18(b) implies that the minimal polynomial of T is z^m for some positive integer m. By 5.29, $m \leq 4$, and since $T^3 e_1 = e_4 \neq 0$, we must have $m \geq 4$, so $m = 4$, and hence the minimal polynomial of T is z^4. ■

Exercise 8B16

Let T be the operator on $\mathbb{C}^6$ defined by

$$T(z_1, z_2, z_3, z_4, z_5, z_6) = (0, z_1, z_2, 0, z_4, 0).$$

Find the characteristic polynomial and the minimal polynomial of T.

Solution. Let $e_1, \ldots, e_6$ be the standard basis of $\mathbb{C}^6$. Observe that $Te_k = e_{k+1}$ for $k = 1, 2, 4$ and $Te_j = 0$ for $j = 3, 5, 6$. Hence, $T^3 = 0$, and T is nilpotent. By 8.17(a), 0 is the only eigenvalue of T, and $\dim \operatorname{null} T^6 = \dim \mathbb{C}^6 = 6$, so the multiplicity of 0 is 6 by 8.23, and hence the characteristic polynomial of T is z^6. Moreover, 8.18(b) implies that the minimal polynomial of T is z^m for some positive integer m. By 5.29, $m \leq 3$, and since $T^2 e_1 = e_3 \neq 0$, we must have $m \geq 3$, so $m = 3$, and hence the minimal polynomial of T is z^3. ■

Exercise 8B17

Suppose $\mathbb{F} = \mathbb{C}$ and $P \in \mathcal{L}(V)$ is such that $P^2 = P$. Prove that the characteristic polynomial of P is $z^m(z-1)^n$, where $m = \dim \operatorname{null} P$ and $n = \dim \operatorname{range} P$.

Solution. Let $d = \dim V$. By induction, $P^d = P$, so $V = \operatorname{null} P^d \oplus \operatorname{range} P^d = \operatorname{null} P \oplus \operatorname{range} P$ by 8.4. Choose a basis of V consisting of a basis of $\operatorname{null} P$ followed by a basis of $\operatorname{range} P$. For any $u \in \operatorname{range} P$, there exists some $w \in V$ such that $Pw = u$. Then $Pu = P(Pw) = P^2 w = Pw = u$, and hence the matrix of P with respect to this basis is diagonal with m zeros followed by n ones. By 5.41, the eigenvalues of P are the diagonal entries, and the multiplicity of each eigenvalue is the number of times it appears on the diagonal by 8.31. Hence, the characteristic polynomial of P is $z^m(z-1)^n$. ■

Exercise 8B18

Suppose $T \in \mathcal{L}(V)$ and λ is an eigenvalue of T. Explain why the following four numbers equal each other.

(a) The exponent of $z - \lambda$ in the factorization of the minimal polynomial of T.

(b) The smallest positive integer m such that $(T - \lambda I)^m|_{G(\lambda,T)} = 0$.

(c) The smallest positive integer m such that

$$\operatorname{null}(T - \lambda I)^m = \operatorname{null}(T - \lambda I)^{m+1}.$$

(d) The smallest positive integer m such that

$$\operatorname{range}(T - \lambda I)^m = \operatorname{range}(T - \lambda I)^{m+1}.$$

Solution. Let $n = \dim V$. Define $S = T - \lambda I$. By 8.20, $G = G(\lambda, T) = \operatorname{null} S^n$. Let the integers in each part be denoted by a, b, c, d respectively. The integer a exists because of 5.27(a); b exists because $S^n|_G = 0$; c exists by 8.3; d exists by Exercise 8A9.

By Exercise 8B6, $a = b$. Applying Exercise 8A9 to S gives $c = d$. We close the proof by showing $b = c$.
$\boxed{\leq}$ By 8.1 and 8.2, $G \subseteq \operatorname{null} S^n \subseteq \operatorname{null} S^{c+n} = \operatorname{null} S^c$, so $b \leq c$.
$\boxed{\geq}$ By 8.1, $G \subseteq \operatorname{null} S^b \subseteq \operatorname{null} S^{b+1} \subseteq G$, so $c \leq b$. Therefore, $b = c$, and hence all four numbers are equal. ■

Exercise 8B19

Suppose $\mathbb{F} = \mathbb{C}$ and $S \in \mathcal{L}(V)$ is a unitary operator. Prove that the constant term in the characteristic polynomial of S has absolute value 1.

Solution. Let $\lambda_1, \ldots, \lambda_k$ be the distinct eigenvalues of S and let $m_j = \dim G(\lambda_j, S)$ for each j. The characteristic polynomial of S has constant term $(-\lambda_1)^{m_1} \cdots (-\lambda_k)^{m_k}$. Since S is unitary, $|\lambda_j| = 1$ for each j by 7.54, so the absolute value of the constant term is $|(-\lambda_1)^{m_1} \cdots (-\lambda_k)^{m_k}| = 1$. ■

Exercise 8B20 ★

Suppose that $\mathbb{F} = \mathbb{C}$ and $V_1, \ldots, V_m$ are nonzero subspaces of V such that

$$V = V_1 \oplus \cdots \oplus V_m.$$

Suppose $T \in \mathcal{L}(V)$ and each V_k is invariant under T. For each k, let p_k denote the characteristic polynomial of $T|_{V_k}$. Prove that the characteristic polynomial of T equals $p_1 \cdots p_m$.

Solution. Let q be the characteristic polynomial of T. For each k, let $n_k = \dim V_k$. Since each V_k is nonzero and complex, 5.47 yields a basis $v_{k,1}, \ldots, v_{k,n_k}$ of V_k with respect to which $\mathcal{M}(T|_{V_k})$ is upper triangular. For each V_k, denote the diagonal entries of $\mathcal{M}(T|_{V_k})$ by $d_{k,j}$ for $j = 1, \ldots, n_k$. Let $\mathcal{B}$ be the list obtained by concatenating the bases of the V_k. By 3.94, the sum of n_k over all k equals $\dim V$. Additionally, $\mathcal{B}$ spans V, so it is a basis of V by 2.42. Because $\mathcal{M}(T|_{V_k})$ is upper triangular for each k, 5.39(c) yields $Tv_{k,j} \in \operatorname{span}(v_{k,1}, \ldots, v_{k,j})$ for each $j = 1, \ldots, n_k$. Moreover,

$\operatorname{span}(v_{k,1},\dots,v_{k,j}) \subseteq \operatorname{span}(v_{1,1},\dots,v_{1,n_1},\dots,v_{k,1},\dots,v_{k,j})$, so $\mathcal{M}(T,\mathcal{B})$ is upper triangular by 5.39(c), where the diagonal entries are precisely the $d_{k,j}$. By 5.41, 8.31, and 8.26 applied to T and $T|_{V_k}$ for each k, we obtain

$$q = \prod_{k=1}^{m}\prod_{j=1}^{n_k}(z-d_{k,j}) = \prod_{k=1}^{m} p_k. \qquad \blacksquare$$

Exercise 8B21

Suppose $p, q \in \mathcal{P}(\mathbb{C})$ are monic polynomials with the same zeros and q is a polynomial multiple of p. Prove that there exists $T \in \mathcal{L}(\mathbb{C}^{\deg q})$ such that the characteristic polynomial of T is q and the minimal polynomial of T is p.

Axler Note. *This exercise implies that every monic polynomial is the characteristic polynomial of some operator.*

Solution. Since $\mathbb{C}^{\deg q} \neq \{0\}$, we have $\deg q \geq 1$, so q has a zero by 4.12. Since p and q have the same zeros, p also has a zero. Write $q = pr$ for some $r \in \mathcal{P}(\mathbb{C})$, and factor r by the fundamental theorem of algebra. By uniqueness of factorization in $\mathcal{P}(\mathbb{C})$, we can write

$$q(z) = \prod_{k=1}^{n}(z-\lambda_k)^{m_k}, \quad p(z) = \prod_{k=1}^{n}(z-\lambda_k)^{\ell_k},$$

where $\lambda_1,\dots,\lambda_n$ are the distinct zeros of q. Moreover, all m_k and ℓ_k are positive integers since p and q have zeros at λ_k, and we have $\ell_k \leq m_k$ for each k since q is a polynomial multiple of p. Note that $\deg q = \sum_{k=1}^{n} m_k$. Let $e_1,\dots,e_{\deg q}$ be the standard basis of $\mathbb{C}^{\deg q}$, and split it into blocks of sizes $m_1,\dots,m_n$ corresponding to the distinct zeros $\lambda_1,\dots,\lambda_n$ of q. For each λ_k, let $v_{k,1},\dots,v_{k,m_k}$ be the ordered subset of the standard basis of $\mathbb{C}^{\deg q}$ as described above. By the linear map lemma, there exists a unique linear map $T \in \mathcal{L}(\mathbb{C}^{\deg q})$ such that

$$Tv_{k,j} = \begin{cases} \lambda_k v_{k,j} + v_{k,j-1}, & \text{for } j = 2,\dots,\ell_k, \\ \lambda_k v_{k,j}, & \text{for } j = 1 \text{ and } j = \ell_k+1,\dots,m_k. \end{cases}$$

Observe that the matrix of T with respect to the standard basis is upper triangular, where λ_k appears on the diagonal m_k times for each k. Therefore, the characteristic polynomial of T is q by 5.41 and 8.31.

Consider the subspaces $V_k = \operatorname{span}(v_{k,1},\dots,v_{k,m_k})$ for each $k = 1,\dots,n$. Then $(T-\lambda_k I)^{\ell_k} v_{k,j} = 0$ for all $j = 1,\dots,m_k$. Therefore, $(T-\lambda_k I)^{\ell_k}|_{V_k} = 0$, but $(T-\lambda_k I)^{\ell_k-1} v_{k,\ell_k} = v_{k,1} \neq 0$. Since $V_k \subseteq G(\lambda_k, T)$, and $\dim V_k = \dim G(\lambda_k, T) = m_k$ by 8.31 and 8.23, we have $V_k = G(\lambda_k, T)$. By 5.27(b), the minimal polynomial of T is a product of the factors $(z-\lambda_k)$, and by Exercise 8B6, the exponents of these factors are exactly ℓ_k. Therefore, the minimal polynomial of T is p. $\blacksquare$

Exercise 8B22

Suppose A and B are block diagonal matrices of the form

$$A = \begin{pmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_m \end{pmatrix}, \quad B = \begin{pmatrix} B_1 & & 0 \\ & \ddots & \\ 0 & & B_m \end{pmatrix},$$

where A_k and B_k are square matrices of the same size for each $k = 1, \dots, m$. Show that AB is a block diagonal matrix of the form

$$AB = \begin{pmatrix} A_1B_1 & & 0 \\ & \ddots & \\ 0 & & A_mB_m \end{pmatrix}.$$

Solution. Let A_k and B_k be n_k-by-n_k matrices for each $k = 1, \dots, m$. Let $N_k = n_1 + \cdots + n_{k-1}$ for each $k = 1, \dots, m$, with the convention that $N_1 = 0$. Then the kth block matrix has indices i, j with $N_k + 1 \le i, j \le N_k + n_k$. Fix ℓ and s with $1 \le s \le n_\ell$, and consider the $(N_\ell + s)$th column of B. The $(N_\ell + s)$th column of B has zeros outside the ℓth block and equals the sth column of B_ℓ within the ℓth block. By 3.48 and 3.50, the $(N_\ell + s)$th column of AB is

$$\sum_{t=1}^{n_\ell} (B_\ell)_{t,s} A_{\cdot, N_\ell + t}.$$

Since A is block diagonal, every column $A_{\cdot, N_\ell + t}$ has zeros outside the rows $N_\ell + 1 \le i \le N_\ell + n_\ell$, and equals $(A_\ell)_{\cdot, t}$ within those rows. Then the $(N_\ell + s)$th column of AB is zero outside the ℓth block and equals

$$\sum_{t=1}^{n_\ell} (B_\ell)_{t,s} (A_\ell)_{\cdot, t} = A_\ell (B_\ell)_{\cdot, s} = (A_\ell B_\ell)_{\cdot, s}$$

by 3.48 and 3.50. Since this holds for any column of any block, we conclude that AB is block diagonal with blocks A_kB_k. ∎

Exercise 8B23

Suppose $\mathbb{F} = \mathbb{R}$, $T \in \mathcal{L}(V)$, and $\lambda \in \mathbb{C}$.

(a) Show that $u + iv \in G(\lambda, T_{\mathbb{C}})$ if and only if $u - iv \in G(\overline{\lambda}, T_{\mathbb{C}})$.

(b) Show that the multiplicity of λ as an eigenvalue of $T_{\mathbb{C}}$ equals the multiplicity of $\overline{\lambda}$ as an eigenvalue of $T_{\mathbb{C}}$.

(c) Use (b) and the result about the sum of the multiplicities (8.25) to show that if $\dim V$ is an odd number, then $T_{\mathbb{C}}$ has a real eigenvalue.

(d) Use (c) and the result about real eigenvalues of $T_{\mathbb{C}}$ (Exercise 5A17) to show that if $\dim V$ is an odd number, then T has an eigenvalue (thus giving an alternative proof of 5.34).

Axler Note. *See* Exercise 3B33 *for the definition of the complexification* $T_{\mathbb{C}}$.

Solution. Define $C : V_{\mathbb{C}} \to V_{\mathbb{C}}$ by $C(u + iv) = u - iv$ for $u, v \in V$. We claim that C is additive, conjugate

linear, satisfies $C^2 = I$, and commutes with $T_{\mathbb{C}}$. To see additivity, let $u_1 + iv_1, u_2 + iv_2 \in V_{\mathbb{C}}$. Then

$$\begin{aligned} C((u_1 + iv_1) + (u_2 + iv_2)) &= C((u_1 + u_2) + i(v_1 + v_2)) \\ &= (u_1 + u_2) - i(v_1 + v_2) \\ &= (u_1 - iv_1) + (u_2 - iv_2) \\ &= C(u_1 + iv_1) + C(u_2 + iv_2). \end{aligned}$$

To see conjugate linearity, let $\alpha = a + bi \in \mathbb{C}$ and $u + iv \in V_{\mathbb{C}}$. Then

$$\begin{aligned} C(\alpha(u + iv)) &= C((a + bi)(u + iv)) \\ &= C((au - bv) + i(av + bu)) \\ &= (au - bv) - i(av + bu) \\ &= (a - bi)(u - iv) \\ &= \overline{\alpha} C(u + iv). \end{aligned}$$

Moreover, $C^2(u + iv) = C(u - iv) = u + iv$, so $C^2 = I$. Finally,

$$CT_{\mathbb{C}}(u + iv) = C(Tu + iTv) = Tu - iTv = T_{\mathbb{C}}(u - iv) = T_{\mathbb{C}}C(u + iv),$$

so C commutes with $T_{\mathbb{C}}$.

(a) Since C commutes with $T_{\mathbb{C}}$ and $C(\lambda(u+iv)) = \overline{\lambda}C(u+iv)$ for any $u+iv \in V_{\mathbb{C}}$, we have $C(T_{\mathbb{C}} - \lambda I) = (T_{\mathbb{C}} - \overline{\lambda} I)C$. Then for any positive integer k, it follows by induction that $C(T_{\mathbb{C}} - \lambda I)^k = (T_{\mathbb{C}} - \overline{\lambda} I)^k C$. By definition, $u + iv \in G(\lambda, T_{\mathbb{C}})$ if and only if $(T_{\mathbb{C}} - \lambda I)^k (u + iv) = 0$ for some positive integer k. Moreover, $C^2 = I$ implies that C is injective, and additivity gives $C0 = 0$, so for any positive integer k, $(T_{\mathbb{C}} - \lambda I)^k (u + iv) = 0$ if and only if $(T_{\mathbb{C}} - \overline{\lambda} I)^k C(u + iv) = 0$. Since $C(u + iv) = u - iv$, we have $(T_{\mathbb{C}} - \overline{\lambda} I)^k (u - iv) = 0$ for some k if and only if $u - iv \in G(\overline{\lambda}, T_{\mathbb{C}})$.

(b) Suppose $v_1, \ldots, v_m$ is a basis of $G(\lambda, T_{\mathbb{C}})$. By part (a), $Cv_j \in G(\overline{\lambda}, T_{\mathbb{C}})$ for each $j = 1, \ldots, m$. We claim that $Cv_1, \ldots, Cv_m$ is a basis of $G(\overline{\lambda}, T_{\mathbb{C}})$. To see this, suppose there are scalars $\alpha_1, \ldots, \alpha_m \in \mathbb{C}$ such that

$$\sum_{j=1}^{m} \alpha_j Cv_j = 0.$$

Applying C to both sides gives

$$C\left(\sum_{j=1}^{m} \alpha_j Cv_j\right) = \sum_{j=1}^{m} \overline{\alpha_j} C^2 v_j = \sum_{j=1}^{m} \overline{\alpha_j} v_j = C0 = 0.$$

By the linear independence of $v_1, \ldots, v_m$, we must have $\overline{\alpha_1} = \cdots = \overline{\alpha_m} = 0$, hence $\alpha_1 = \cdots = \alpha_m = 0$. Moreover, for any $w \in G(\overline{\lambda}, T_{\mathbb{C}})$, we have $Cw \in G(\lambda, T_{\mathbb{C}})$ by part (a), so $Cw = \sum_{j=1}^{m} \beta_j v_j$ for some $\beta_j \in \mathbb{C}$, hence

$$w = C(Cw) = C\left(\sum_{j=1}^{m} \beta_j v_j\right) = \sum_{j=1}^{m} \overline{\beta_j} Cv_j.$$

Therefore, $Cv_1, \ldots, Cv_m$ is a basis of $G(\overline{\lambda}, T_{\mathbb{C}})$, whence $\dim G(\overline{\lambda}, T_{\mathbb{C}}) = \dim G(\lambda, T_{\mathbb{C}})$. Thus, λ is an eigenvalue of $T_{\mathbb{C}}$ if and only if $\overline{\lambda}$ is an eigenvalue of $T_{\mathbb{C}}$.

(c) By Exercise 2B11, $\dim V_{\mathbb{C}} = \dim V$. Then $V_{\mathbb{C}}$ is nonzero with odd dimension, so we may apply 8.25 to $T_{\mathbb{C}}$ to conclude that $\dim V_{\mathbb{C}} = \sum_{\mu} d_{\mu}$, where μ ranges over the eigenvalues of $T_{\mathbb{C}}$ and d_{μ} is the multiplicity of μ. Note that for nonreal eigenvalues μ, we have $\mu \neq \overline{\mu}$. By part (b), $d_{\mu} = d_{\overline{\mu}}$ for each nonreal eigenvalue μ, so the sum of the multiplicities of nonreal eigenvalues is even. Since $\dim V_{\mathbb{C}}$ is odd, there must be at least one real eigenvalue of $T_{\mathbb{C}}$.

(d) By (c), $T_{\mathbb{C}}$ has a real eigenvalue $\nu \in \mathbb{R}$. By Exercise 5A17, ν is an eigenvalue of T. ■